Modern Models, Historical isiXhosa: A Held-Out Study of Language-Modelling Fit

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Abstract

Language coverage does not establish coverage of a language's historical varieties. We evaluate five frozen pretrained models on nineteenth-century isiXhosa newspapers and a modern NCHLT comparison corpus. Final evaluation uses 300 sentences per corpus from disjoint newspaper issues and modern documents, with character-length stratification. XLM-R, Afro-XLMR and InkubaLM assign worse character-normalised scores to historical text, with median gaps of 0.796, 0.815 and 0.627 and source-group bootstrap intervals excluding zero. mBERT and AfriBERTa show no clear directional difference. Validation reproduces the same pattern. These findings indicate a historical-modern corpus mismatch for three models; they do not demonstrate semantic failure or isolate temporal change from register, orthography and transcription differences.

Introduction

The IsiXhosa Intellectual Traditions Digital Archive preserves writing that is central to studying African intellectual history (Schoots et al., 2023). Making this material digitally accessible, however, does not establish that present-day NLP systems process it adequately. We ask: do selected modern pretrained models assign worse language-modelling scores to isiXhosa newspaper text from 1870-1894 than to a modern comparison corpus? We hypothesise a positive historical-modern scoring gap in models with documented isiXhosa training coverage.

Historical English research finds benefits from historically specific pretraining (Manjavacas & Fonteyn, 2022), motivating evaluation across periods rather than assuming transfer. African-language adaptation likewise improves selected downstream benchmarks (Alabi et al., 2022), while AfriBERTa demonstrates the value of targeted multilingual pretraining (Ogueji et al., 2021). These studies motivate our experiment but do not establish historical isiXhosa performance. We use pseudo-log-likelihood (PLL) for masked models (Salazar et al., 2020) and causal log-likelihood for InkubaLM. The experiment measures distributional fit, not semantic understanding or whether a model identifies the text as a foreign dialect.

Methods

The historical input comprises the supplied isiXhosa text blocks from Imvo Zabantsundu and Isigidimi sama-Xosa. The tracker covers 3,608 pages; a 100-isiXhosa-word minimum retains 3,592. Automated normalisation, segmentation and filtering yield 327,265 sentences from 3,574 pages. Sentences require at least three whitespace words and 25 characters, with at most 80 words. A proposed 90% page-level isiXhosa word-share filter was rejected: the diagnostic showed it would additionally exclude 895 substantial pages, although their English blocks were already separated. Figure 2 shows the diagnostic thresholds. The spike at 3,000 reflects display clipping, not an exact page word count. Modern comparison text is the NCHLT clean isiXhosa corpus (Puttkammer et al., 2014), associated with text-resource development for South African languages (Eiselen & Puttkammer, 2014). After automated filtering, duplicate removal and a 200-sentence document cap, 41,188 sentences remain. Government-domain provenance (Puttkammer et al., 2014) makes this a register-mismatched comparator. We performed no manual correction or modernisation because we lacked isiXhosa proficiency. Automated processing is distinct from manual linguistic verification. After exploratory scoring, we created source-disjoint development, validation and test partitions. All pages of a historical issue share a partition; modern grouping uses original document identifiers. Previously evaluated sources are assigned to development. Normalised exact duplicates and exact cross-corpus overlaps are removed. The resulting pools contain 263,748 historical and 41,188 modern sentences. Validation and test each contain 300 sentences per corpus, sampled equally across five character-length bins defined from development historical text (seed 20261031). Validation covers 100 historical issues and 114 modern documents; test covers 101 and 112. Models remain frozen: the development partition is not used for training or hyperparameter optimisation. The checkpoints are bert-base-multilingual-cased, castorini/afriberta_large, FacebookAI/xlm-roberta-large, Davlan/afro-xlmr-large and lelapa/InkubaLM-0.4B. XLM-R uses multilingual web pretraining (Conneau et al., 2020); the Afro-XLMR and InkubaLM documentation explicitly includes isiXhosa (Alabi et al., 2022) (Tonja et al., 2024). mBERT and AfriBERTa serve as comparison baselines, without isiXhosa in their documented language lists (Ogueji et al., 2021) (Devlin et al., 2018). For each sentence, we divide total negative log score by its character count. Masked models score each non-special subword while other tokens remain visible; the causal model uses shifted next-token loss. The primary statistic is the historical minus modern median within each model. We report 95% percentile intervals from 2,000 source-group bootstrap draws, Cliff's delta, and approximate centred-bootstrap p-values with Holm correction across the five models.

Results

The held-out test reproduces validation (Table 1; Figures 1 and 3). All three positive-gap model intervals exclude zero; approximate Holm-adjusted bootstrap p-values are 0.0025. Both baseline intervals include zero. Test mean lengths are closely aligned: 111.41 historical versus 111.47 modern characters (medians 91 and 100). Figure 4 documents the

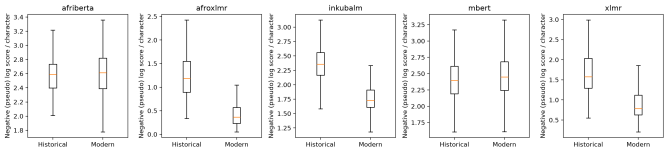


Figure 1: Final held-out score distributions for 300 historical and 300 modern isiXhosa sentences. Lower character-normalised negative (pseudo) log scores indicate better fit within each model. XLM-R, Afro-XLMR and InkubaLM show higher historical medians, while mBERT and AfriBERTa show substantial overlap between corpora. Panels use different scales; absolute scores should not be used to rank models. Outlier markers are omitted.

original exploratory comparison. [1]

Table 1: Historical minus modern median negative (pseudo) log score per character. Positive gaps indicate worse historical fit within a model. Test: 300 sentences per corpus. Intervals resample source groups; delta is a descriptive sentence-order effect. Absolute scores and gap magnitudes are not a common performance ranking across masked and causal objectives.

Model	Validation gap	Test gap (95% CI)	Cliff's delta
mBERT	-0.039	-0.056 [-0.128, 0.024]	-0.093
AfriBERTa	-0.060	-0.026 [-0.067, 0.020]	-0.047
XLM-R	0.857	0.796 [0.696, 0.872]	0.669
Afro-XLMR	0.842	0.815 [0.725, 0.885]	0.837
InkubaLM	0.629	0.627 [0.578, 0.682]	0.825

Discussion

The stable direction across validation and test supports a corpus-level historical disadvantage for XLM-R, Afro-XLMR and InkubaLM under the chosen objectives. It does not show that these models cannot represent archival isiXhosa meaningfully. Nor does the baseline pattern establish historical robustness: a small difference can arise when neither corpus is well modelled. The small AfriBERTa reverse difference from the initial exploration is not clearly reproduced in either held-out partition. Language inclusion alone therefore should not be treated as sufficient evidence of historical coverage, and these five checkpoints do not identify a causal effect of pretraining language composition. Earlier exploratory comparisons found higher character-normalised negative log scores, indicating poorer fit, for Imvo than Isigidimi in all five models. This is a newspaper contrast, not evidence that progressively older language is harder: publication period, content and text quality overlap. Similarly, NCHLT government text and historical newspapers differ in register and topic. Orthographic conventions, residual errors, unknown pretraining overlap and imperfect automated sentence boundaries could contribute to the main gap. Character normalisation reduces dependence on token counts but does not eliminate tokenizer effects. Standard subword PLL can inflate scores when other pieces of the target word remain visible (Kauf & Ivanova, 2023). The retained objective constrains interpretation. AfriBERTa also shows metric sensitivity: its secondary per-token test gap is 0.223 [0.045, 0.361], unlike its near-zero primary per-character gap. An exploratory synthetic-noise check on 100 modern sentences found that nominal corruption of 1%, 2% and 5% of letters produced score shifts equivalent to approximately 18-24%, 38-46% and 79-87% of the original mean per-token gaps in the three positive-gap models. These are sensitivity comparisons under a generic corruption mechanism, not measured percentages of historical degradation caused by OCR. They use the exploratory sample and a different summary statistic from Table 1. Actual residual transcription error and its score effect remain unmeasured without scan-verified reference passages. For responsible deployment, archival coverage warrants direct evaluation rather than inference from a language label. Poorer modelling fit could affect archival retrieval, classification or other access tools, but no downstream harm was measured here. Researchers, heritage institutions and isiXhosa-speaking communities could be affected if such systems privilege contemporary forms. Conversely, aggressively removing unusual forms or modernising spelling without expertise risks discarding historical evidence. Appropriate next steps are speaker- and historian-informed evaluation, register-matched modern comparisons, and verified transcription audits. Historical-domain adaptation is a research direction suggested by prior work, not an intervention tested in this project.

Conclusion

Three evaluated models assign consistently worse language-modelling scores to the sampled nineteenth-century isiXhosa corpus than to modern NCHLT text, including on a source-disjoint final test. The result supports a historical-modern corpus mismatch while leaving its linguistic and technical causes unresolved. It motivates historically informed evaluation; it does not establish semantic failure, dialect classification or measured downstream exclusion.

Contribution Statement

Kaylin Aspoas (3163666) led the construction and cleaning of the historical corpus, implemented the language-model evaluation pipeline and conducted robustness, and produced the project's visualisations. Lisa Pitsi (2590083) led the research framing and Responsible NLP motivation for the project, developed the modern comparison corpus from NCHLT resources, and performed the statistical comparison between historical and modern corpora. Both authors jointly interpreted the results, discussed their implications, and collaborated on writing, editing, and refining the final report and extended abstract.

[1] Code and results: https://github.com/lisabeeaan/NLP-Project2

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Supplementary Figures

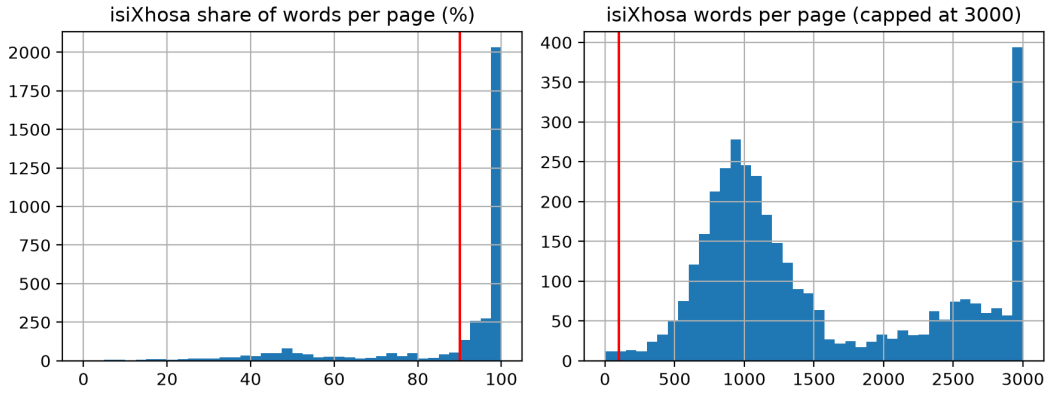


Figure 2: The left panel shows that most pages have a high isiXhosa word share, while a substantial minority contain more English elsewhere on the page. Its red 90% line is a rejected diagnostic cutoff, not the active filter. The right red line marks the retained 100-isiXhosa-word minimum. Values above 3,000 are clipped for display, so the final spike does not mean those pages contain exactly 3,000 words.

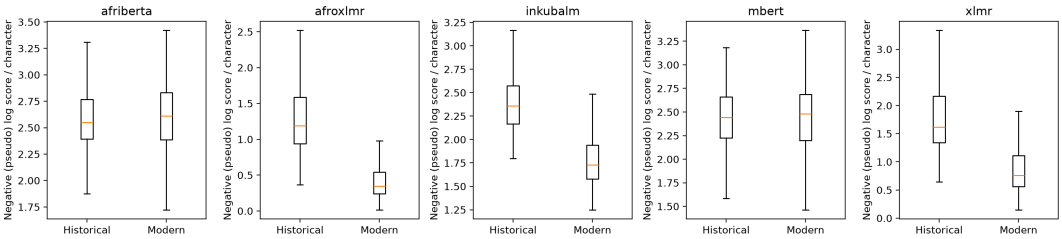


Figure 3: Validation shows the same directional pattern as the final test: a historical disadvantage for XLM-R, Afro-XLMR and InkubaLM and substantial overlap for the two baselines. Agreement across separate source groups supports the stability of the main corpus comparison. It does not isolate temporal language change from other corpus differences.

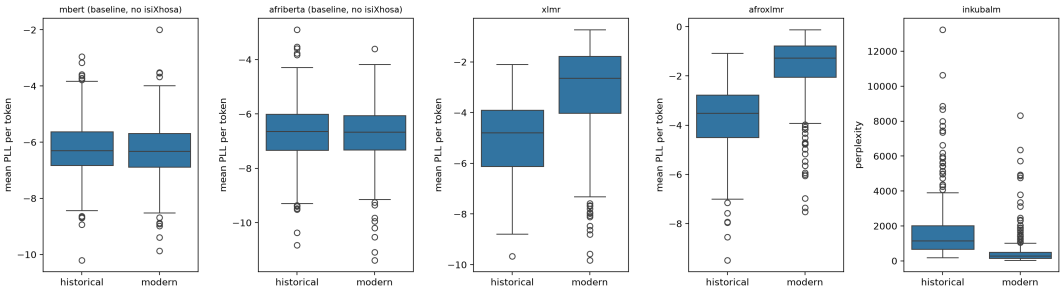


Figure 4: The original sample already shows worse historical scores for XLM-R, Afro-XLMR and InkubaLM, with close baseline medians. For the four masked models, higher mean PLL per token is better, so a lower historical box indicates poorer fit. InkubaLM uses perplexity, where lower is better, so its higher historical box indicates poorer fit. These are exploratory results on the original sample, not the final character-normalised held-out results.

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